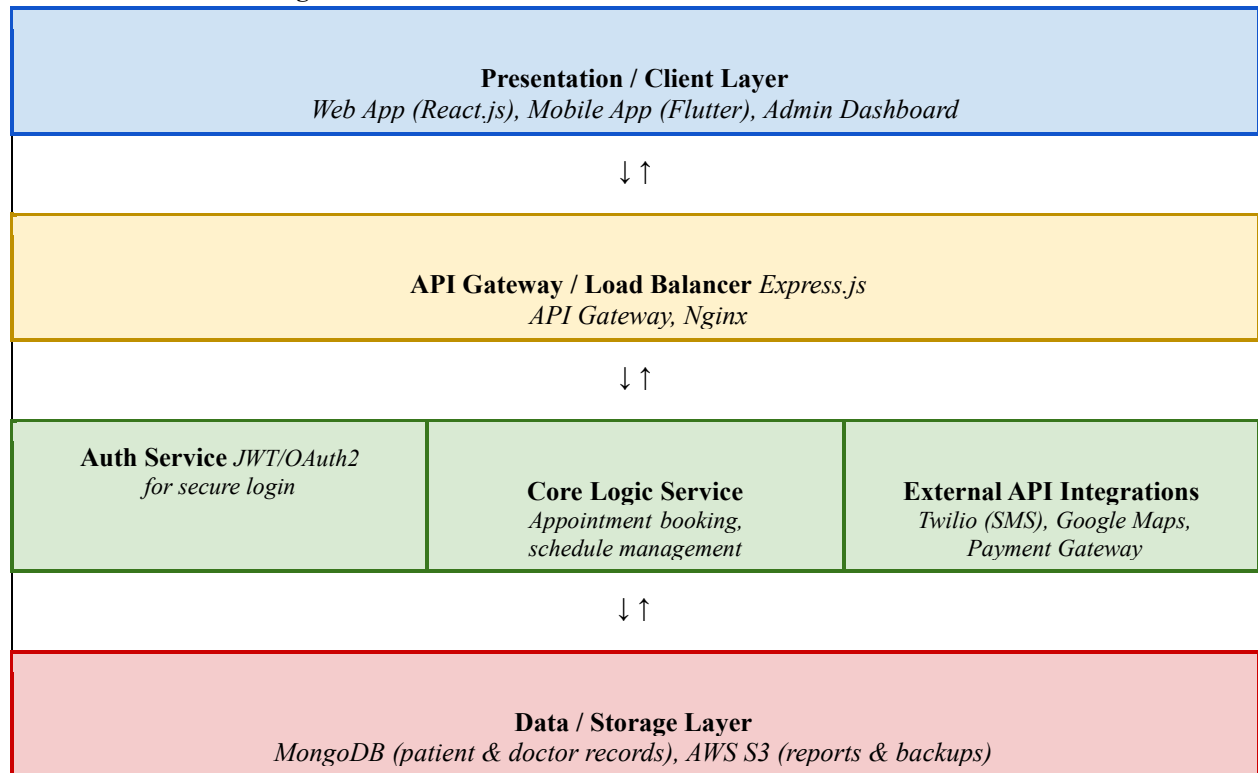


## Solution Architecture

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 15 July 2026  |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Book a Doctor |
| <b>Maximum Marks</b> | 5 Marks       |

### Solution Architecture Diagram



### Layered View Summary

| Layer                       | Components                                                |
|-----------------------------|-----------------------------------------------------------|
| Presentation / Client Layer | Web App (React.js), Mobile App (Flutter), Admin Dashboard |
| API Gateway / Load Balancer | Express.js API Gateway, Nginx                             |

|                                   |                                                                                                                                                                                                                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application Layer (Microservices) | <ul style="list-style-type: none"> <li>- Auth Service: JWT/OAuth2 for secure login</li> <li>- Core Logic Service: Appointment booking, schedule management -</li> </ul> <p>External API Integrations: Twilio (SMS reminders), Google Maps (clinic location), Payment Gateway</p> |
| Data / Storage Layer              | MongoDB (patient & doctor records), AWS S3 (reports & backups)                                                                                                                                                                                                                   |

## Component Description Table

| Component Name           | Description / Role in Architecture                                                                      | Technologies Used                         |
|--------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Presentation Layer       | Provides user interface for patients (booking), doctors (schedule management), and admins (monitoring). | React.js, Flutter, Bootstrap              |
| API Gateway              | Routes requests between client and backend services, ensures load balancing and security.               | Express.js, Nginx                         |
| Microservices            | Handles authentication, appointment booking, notifications, and payment processing.                     | Node.js, JWT, Twilio API, Google Maps API |
| Database / Storage Layer | Stores patient records, doctor schedules, and transaction logs securely.                                | MongoDB, AWS S3                           |